

REAL WEATHER AUDIT

StormWise DI² Belmar

Cloud-Cover Decision Audit

Pre-Event Forecast Versus Observed Beach Conditions

Belmar, New Jersey — July 31, 2026

Audit type	Pre-event forecast versus observed beach conditions	Decision boundary	Below 60% = GO
Archived issuance	July 30, 2026 — 10:23 PM EDT	At or above	60% = NO-GO
Evidence posture	Source-bound canonical report	Rule layer	Tier-18 weather-audit posture

FINAL AUDIT STATUS

OPERATIONAL DECISION VERIFIED — PEAK TIMING SUPPORTED — EXACT PEAK PARTIALLY VERIFIED

Prepared from the supplied archived forecast, station chronology, beach photographs, observer reports, commercial displays, webcam comparison, and derivative audit record.

INTRODUCTION

Introduction: Decision Context

This audit began with an ordinary question: would Belmar Beach be sunny enough to justify the trip? The participants were not evaluating emergency conditions or attempting to establish a universal forecasting claim. They simply preferred a beach day with meaningful sunshine and wanted a clear rule for making the decision.

As a practical demonstration, they agreed in advance to use 60% cloud cover as the decision boundary:

- Below 60% cloud cover: GO
- At or above 60% cloud cover: NO-GO

That threshold converted a subjective preference—“we want enough sun”—into a testable operational decision. Several mainstream products suggested conditions near or above the NO-GO boundary, while the archived StormWise DI² analysis placed the likely peak below it and anticipated clearing later in the afternoon.

The DI Weather Station analysis was not rerun on the morning of the event. Even so, the prior-night DI² forecast materially changed the participants' decision posture: instead of treating the cloudier commercial outlook as a reason to cancel, they retained the trip as a GO and later compared the forecast with direct beach observations.

That distinction matters. Without the earlier DI² analysis—and without later access to on-site photographs, observer reports, station history, or beach webcams—the participants might reasonably have relied on the more pessimistic commercial displays and stayed home. Had they done so, they may never have discovered that the decision was inconsistent with their actual goal: finding a beach period with substantial sunshine and cloud cover below their chosen threshold.

The event therefore provides a useful small-scale demonstration of decision value. The issue was mundane, but the test was concrete: could a pre-event forecast place the conditions on the correct side of a practical threshold, early enough to change behavior?

For this event, that is the question the audit evaluates.

1. Executive Summary

StormWise DI² issued a pre-event GO / beach-specific conditional GO forecast at 10:23 PM EDT on Thursday, July 30, 2026. The governing operational rule was:

Cloud cover below 60% = GO

Cloud cover at or above 60% = NO-GO

The archived DI² forecast called for an early-afternoon cloud maximum of approximately 52% near 1:30 PM, followed by clearing. It rejected the sharper 60–68% mostly cloudy scenario as the leading outcome.

The event record supports the practical decision. Station history changed from Fair at 12:56 PM to Mostly Cloudy at 1:56 PM, consistent with a temporary early-afternoon increase in clouds. By 2:43 PM, direct beach photographs showed substantial blue sky and uneven cumulus, while the on-site observer estimated approximately 30% cloud cover. At 2:50 PM, the observer reported that cloud cover had been near 40% on arrival and had cleared during the preceding hour.

At approximately 3:00–3:01 PM, AccuWeather displayed 51% cloud cover and intermittent clouds, while another commercial product displayed 60% “now” and 66% at 4 PM. A simultaneous ocean-facing beach webcam appeared much sunnier than the latter product suggested, although the webcam did not show the entire sky dome.

Event-specific determination

- Operational GO classification: VERIFIED
- Early-afternoon cloud maximum: SUPPORTED
- Clearing after the maximum: VERIFIED
- Exact 52% peak magnitude: PARTIALLY VERIFIED
- Literal proof that the peak never reached 60%: UNRESOLVED
- Visible beachfront conditions below 60% by 2:43–3:01 PM: VERIFIED

The strongest defensible conclusion is that DI² correctly made the practical GO decision and correctly identified the cloud-growth-then-clearing pattern. The exact peak cloud fraction was not directly measured by a calibrated whole-sky instrument.

OPERATIONAL CONCLUSION

GO classification verified; peak timing supported; exact numerical peak unresolved.

2. Audit Scope

Geographic scope

The audit concerns Belmar Beach and the immediate Belmar, New Jersey area. Evidence includes:

- a nearby station-history display;
- direct photographs taken on the beach;
- contemporaneous observer reports;
- a Belmar ocean-facing webcam;
- commercial cloud-cover products.

Temporal scope

The evaluated target period was 10:00 AM–6:00 PM EDT on Friday, July 31, 2026.

The relevant evidence period begins with the archived forecast at 10:23 PM on July 30 and extends through the commercial-product and webcam comparison at approximately 3:01 PM on July 31.

Primary endpoint

Whether the event was correctly classified as:

- GO: cloud cover below 60%;
- NO-GO: cloud cover at or above 60%.

Secondary endpoints

- Timing of the afternoon cloud maximum.
- Magnitude of the maximum.
- Occurrence of later clearing.
- Accuracy of the practical beach-use decision.
- Representativeness of commercial cloud-cover products.

3. Pre-Event Forecast Record

Archived issuance

Thursday, July 30, 2026 — 10:23 PM EDT

The preserved DI² forecast stated:

- Final peak call: 52% at approximately 1:30 PM EDT.
- 10 AM–2 PM: partly cloudy.
- 3 PM: mixed/intermittent.
- 4–6 PM: clear/mostly sunny.
- Operational verdict: GO.
- Beach-specific verdict: conditional GO.
- NWS reference state: approximately 42–43% cloud cover through early afternoon, declining later.
- A sharp 60–68% cloud maximum was not accepted as the leading scenario.
- Late-afternoon clearing was expected.

The forecast was not subsequently rewritten to conform to the observed outcome.

Lead times

The archived forecast preceded:

- the predicted 1:30 PM peak by 15 hours 7 minutes;
- the 2:43 PM beach photographs by 16 hours 20 minutes;
- the 3:01 PM commercial-product comparison by 16 hours 38 minutes.

4. Timestamped Evidence Timeline

Evidence classes are preserved as supplied. Forecasts, categorical station displays, direct observations, modeled products, webcam evidence, and derivative analysis are not treated as interchangeable.

Date / time EDT	Class / source	Condition or displayed value	Relevance	Limitation
Jul. 30 / 10:23 PM	A / Archived DI ² forecast	52% peak near 1:30 PM; under-60 GO; clearing later	Establishes pre-event call and decision threshold	Forecast, not observation
Jul. 31 / 12:56 PM	B / Station-history display	Fair	Supports relatively open sky shortly before the predicted peak	Categorical only; no exact cloud percentage
Jul. 31 / 1:56 PM	B / Station-history display	Mostly Cloudy	Supports a temporary cloud increase near the predicted peak period	Does not establish exact percentage; earlier displayed 0°F fields appear invalid
Jul. 31 / 2:43 PM	C / Two beach photographs	Substantial blue sky with uneven cumulus	Direct visual evidence of the beachfront sky	Limited directions; not whole-sky calibrated
Jul. 31 / 2:43 PM	D / On-site observer	"I'd say about 30%. Not even in all directions."	Contemporaneous estimate of visible sky cover	Human estimate, not instrument measurement
Jul. 31 / 2:50 PM	D / On-site observer	Approximately 40% on arrival; clearing during prior hour	Supports a higher earlier value followed by clearing	Approximate recollection
Jul. 31 / 3:00 PM	E / AccuWeather display	51%, intermittent clouds	Numerically close to DI ² 's 52% estimate	Modeled/grid product; later than forecast peak
Jul. 31 / 3:01 PM	E / Weather Channel display	60% now; 66% at 4 PM; 59% at 5 PM; 27% at 6 PM	Shows a cloudier commercial-product scenario	Modeled product, not direct beach measurement
Jul. 31 / ~3:01 PM	F / Belmar beach webcam	Predominantly blue, sunny ocean-facing sector	Represents visible beachfront experience	Restricted field of view
Jul. 31 / 3:04 PM	G / DI ² comparison screenshot	Preserves later analytical comparison	Documents audit continuity	Derivative analysis; not independent evidence

5. Forecast-to-Observation Comparison

DI² peak estimate: 52% near 1:30 PM

No supplied instrument directly measured total hemispheric cloud fraction at 1:30 PM. The precise 52% value therefore cannot be exactly verified.

The supporting evidence is indirect:

- 12:56 PM: Fair
- 1:56 PM: Mostly Cloudy
- 2:50 PM observer report: conditions had cleared during the previous hour
- 3 PM AccuWeather product: 51%

Together, those records are consistent with cloud cover rising during the early afternoon and then declining. They do not prove that the maximum was exactly 52%.

Under-60 operational classification

By 2:43–2:50 PM, the direct photographs and observer estimates placed visible cloud cover around 30–40%, below the operational boundary.

AccuWeather's 3 PM value was also below the threshold at 51%.

The station's 1:56 PM "Mostly Cloudy" category prevents a conclusive finding that the total-sky peak itself never reached 60%, because that category was not accompanied by a precise percentage.

Clearing trend

The clearing pattern is supported by multiple distinct evidence classes:

- the observer's 2:50 PM statement that it had cleared during the prior hour;
- the substantial blue sky in the 2:43 PM photographs;
- the much clearer ocean-facing webcam by approximately 3 PM;
- the commercial products' later reductions, including 27% by 6 PM in one display.

Commercial-product divergence

At approximately 3 PM:

- AccuWeather: 51%, intermittent clouds.
- Weather Channel: 60% now, increasing to 66% at 4 PM.
- Observer: approximately 30–40%.
- Webcam: predominantly blue ocean-facing sky.

The commercial products were not equivalent. AccuWeather was numerically closer to the DI² pre-event call, while the 60–66% product did not represent the visible beachfront experience well at the time of comparison.

6. Threshold Determination

Operational decision

DETERMINATION

VERIFIED

The totality of direct beach evidence supports the conclusion that conditions were suitable for the planned outdoor beach activity and were visibly below the 60% NO-GO boundary by the principal verification period.

The GO decision was not dependent on predicting an exact cloud fraction. It depended on placing the event in the correct operational class.

Literal peak under 60%

DETERMINATION

PARTIALLY VERIFIED / UNRESOLVED AT THE EXACT PEAK

The available evidence does not include a calibrated whole-sky cloud measurement during the probable 1:30–2:00 PM maximum. The station’s “Mostly Cloudy” label confirms a temporary cloudier interval but does not supply a numerical fraction.

Accordingly:

- Threshold decision outcome: verified.
- Exact proof that peak total-sky cloud fraction remained below 60%: unresolved.

7. Peak-Cloud Determination

Peak timing

DETERMINATION

SUPPORTED

The change from Fair at 12:56 PM to Mostly Cloudy at 1:56 PM, followed by reported clearing before 2:43–2:50 PM, supports a maximum within approximately 1:30–2:00 PM.

Peak magnitude

DETERMINATION

PARTIALLY VERIFIED

The exact 52% value was not directly measured.

It was nevertheless consistent with:

- a temporary station classification of “Mostly Cloudy”;
- the later AccuWeather value of 51%;
- the observer’s report that conditions had been somewhat cloudier earlier;
- the absence of evidence for a prolonged, opaque overcast deck.

The AccuWeather 51% value is corroborative only. It was a later modeled value and cannot be treated as a direct measurement of the 1:30 PM peak.

Later clearing

DETERMINATION

VERIFIED

The observer’s statement, the 2:43 PM photographs, and the approximately 3 PM webcam collectively establish that conditions became visibly clearer after the early-afternoon cloudier interval.

8. Commercial Product Performance

AccuWeather

AccuWeather displayed 51% at 3 PM with “intermittent clouds.”

That value was numerically one percentage point below DI²’s pre-event 52% peak estimate. The comparison is noteworthy but not a direct verification of the earlier peak because:

- the timestamp was later;
- the number was modeled rather than instrument-measured;
- local cloud cover was changing over time.

AccuWeather’s qualitative description was reasonably compatible with mixed sun and clouds.

Weather Channel

Weather Channel displayed:

- 60% at 3:01 PM;
- 66% at 4 PM;
- 59% at 5 PM;
- 27% at 6 PM.

Its 3:01 PM and 4 PM values did not match the visible beachfront experience shown by the photographs, observer reports, and ocean-facing webcam.

Potential explanations include, as interpretation rather than established fact:

- a broader spatial grid than the beach point;
- delayed model updating;
- clouds concentrated inland or outside the webcam’s field;
- multilayer cloud accounting;
- temporal smoothing of a rapidly changing cloud field.

There is no evidence that the service deliberately ignored local data. The supportable conclusion is only that its displayed value did not represent the contemporaneous visible beach conditions well.

Direct evidence versus modeled percentages

The most accurate source for the perceived beachfront experience was the combination of:

- direct photographs;
- contemporaneous observer reports;
- the live beach webcam.

These sources were limited for measuring total hemispheric cloud fraction but were strongest for answering whether the beach was sunny and usable.

9. Decision Value

The forecast was designed to support a binary decision, not to establish an instrument-grade cloud climatology.

A forecast of 52% and an observation of 40% can differ numerically while producing the same correct decision: GO.

Conversely, a 60–66% modeled value can be numerically close to a threshold but produce a different practical recommendation: NO-GO.

For this event, the most consequential result was that DI² placed the conditions on the usable side of the threshold and anticipated clearing. That decision was supported by the observed beach experience.

DECISION-VALUE FINDING

The operational classification, not an instrument-grade peak number, was the consequential endpoint.

10. Findings Table

Item	Determination	Basis
Operational GO/NO-GO classification	VERIFIED	Beach remained usable; direct evidence showed substantial sun and blue sky
Under-60% threshold during direct verification period	VERIFIED	Observer estimated 30–40%; AccuWeather displayed 51%
Under-60% status at the exact peak	UNRESOLVED	No calibrated whole-sky percentage; station only said “Mostly Cloudy”
Peak timing near 1:30–2:00 PM	SUPPORTED	Fair at 12:56, Mostly Cloudy at 1:56, clearing before 2:43–2:50
Exact 52% peak magnitude	PARTIALLY VERIFIED	Consistent with chronology and later 51% modeled value, but not directly measured
Clearing trend	VERIFIED	Observer statement, photographs, webcam, and later product decline
On-site beach usability	VERIFIED	Direct photographs and reports showed a sunny, usable beach
AccuWeather representation	SUPPORTED	51% and intermittent clouds broadly fit the mixed-sky scenario
Weather Channel representation at ~3 PM	NOT VERIFIED as locally representative	60–66% did not match visible beachfront experience
Archive integrity	VERIFIED	Forecast screenshot preceded the observations and retained its original language

11. Limitations

- No calibrated whole-sky camera was supplied.
- No ceilometer-derived total cloud fraction was supplied.
- The beach photographs show only selected directions.
- The webcam shows an ocean-facing sector rather than the full sky dome.
- Human cloud-cover estimates are approximate.
- Commercial percentages are modeled or gridded products, not direct cloud-fraction measurements.
- Station labels such as “Fair” and “Mostly Cloudy” are categorical.
- The exact 1:30 PM sky fraction was not instrumentally recorded.
- Invalid or missing early temperature and dew-point entries in the station screenshot were excluded.
- One event cannot demonstrate universal forecasting superiority or long-term calibration.

SCOPE RESTRAINT

The report does not establish that total cloud cover remained below 60% at every moment, nor does it establish general superiority across future events.

12. Final Determination

StormWise DI² correctly classified the Belmar Beach event as an operational GO under the prespecified cloud-cover decision framework.

The forecast's early-afternoon cloud maximum and subsequent clearing were supported by the station chronology and observer reports. The transition from Fair at 12:56 PM to Mostly Cloudy at 1:56 PM, followed by visibly clearer conditions before 2:43–2:50 PM, supports the predicted timing pattern.

The exact 52% peak was not directly measured. It was, however, compatible with the temporary cloudier station classification and was numerically close to AccuWeather's later 51% product value.

By approximately 2:43–3:01 PM, direct beach observations showed substantially clearer conditions than the 60–66% values displayed by another commercial product.

Final audit status

STATUS

The event verifies the operational GO decision, supports the peak timing and clearing pattern, and only partially verifies the exact numerical peak.

It does not establish that total cloud cover remained below 60% at every moment, nor does it establish general superiority across future events.

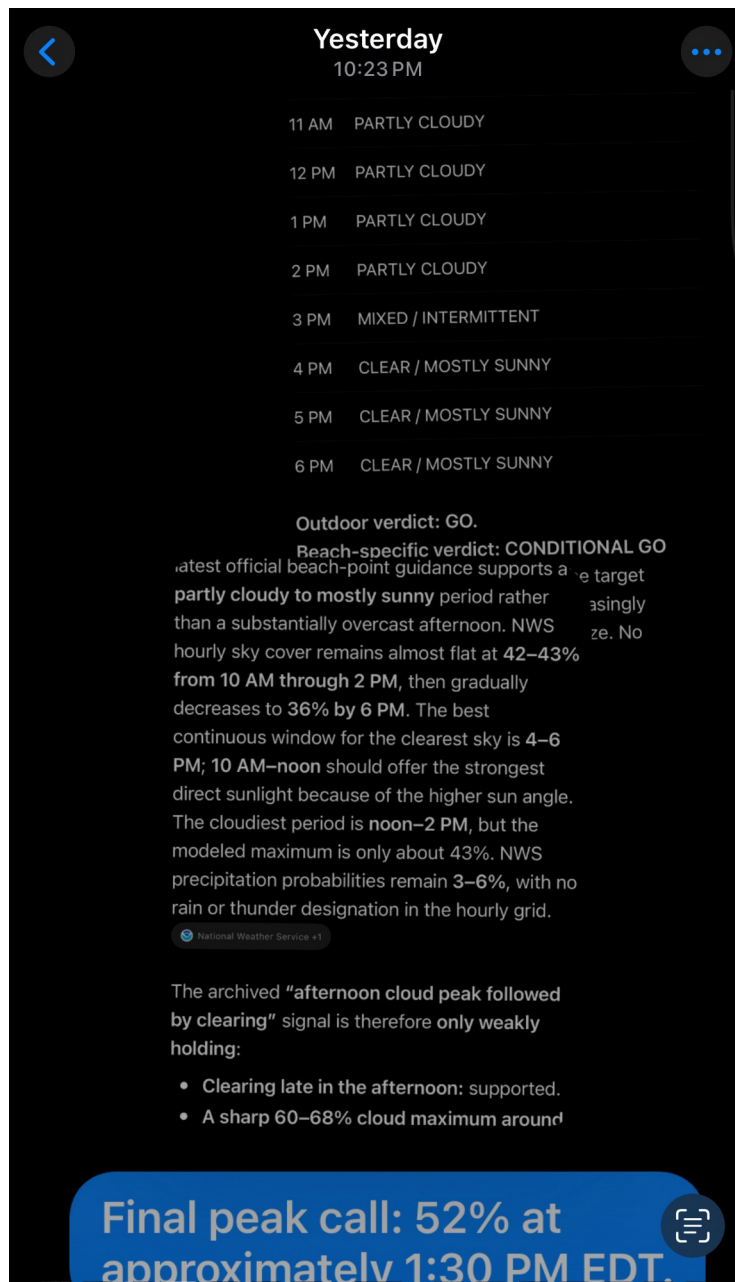
13. Evidence Plates

The plates preserve the supplied screenshots as source-bound evidence. Captions identify the evidentiary role and the limitations stated in the audit record.

EVIDENCE PLATE 01

Plate 1 — Archived DI² Forecast

Screenshot visibly marked “Yesterday 10:23 PM,” preserving the pre-event classification of partly cloudy through 2 PM, mixed/intermittent at 3 PM, and clear/mostly sunny from 4–6 PM, together with the 52% peak call.



EVIDENCE HANDLING

Forecast evidence. It establishes the archived pre-event call and is not treated as an observation.

EVIDENCE PLATE 02

Plate 2 — Station-History Chronology

Daily-observation screenshot showing “Fair” at 12:56 PM and “Mostly Cloudy” at 1:56 PM. Earlier 0°F fields appear invalid and were not used.



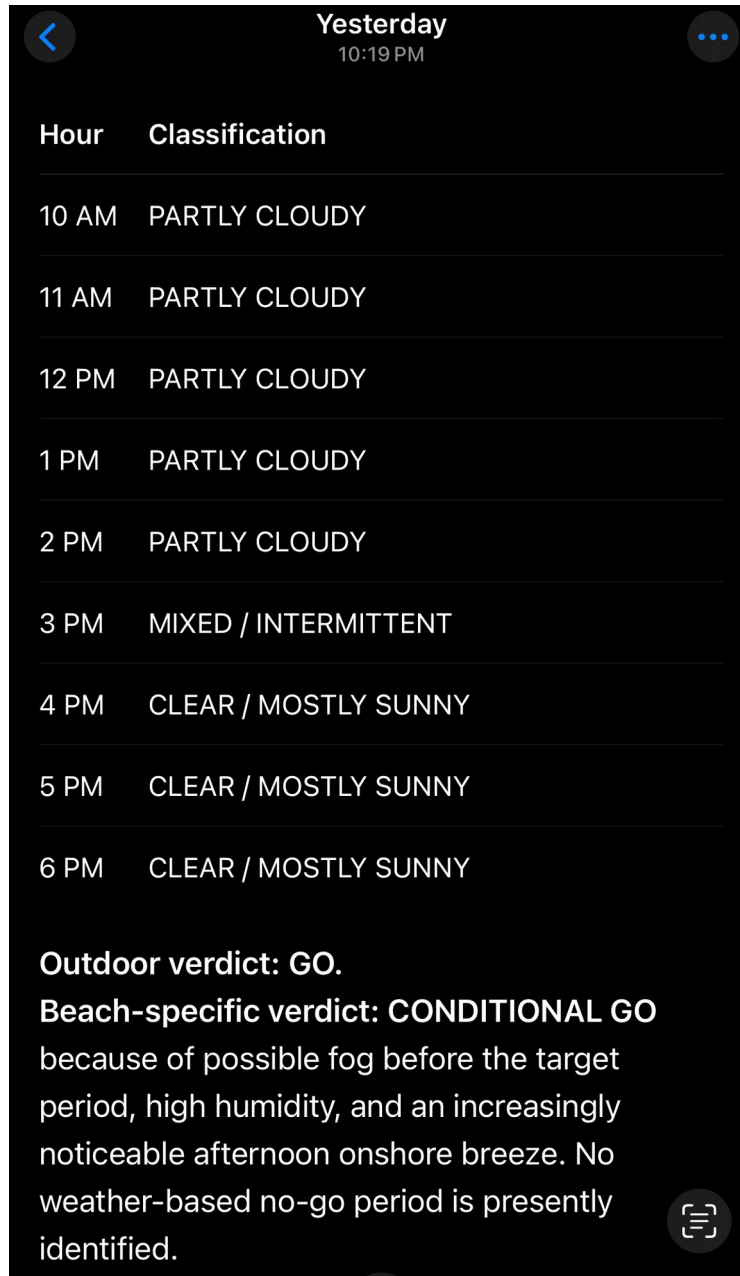
EVIDENCE HANDLING

The supplied capture is retained as a weather-display plate; the categorical station chronology is preserved in the audit record.

EVIDENCE PLATE 03

Plate 3 — Archived Hourly Classification Companion

The preserved hourly classification record shows 10 AM–2 PM as partly cloudy, 3 PM as mixed/intermittent, and 4–6 PM as clear/mostly sunny, with the outdoor verdict GO and beach-specific verdict CONDITIONAL GO.



The screenshot shows a mobile app interface for 'Yesterday' at 10:19 PM. It features a table with two columns: 'Hour' and 'Classification'. The data rows show weather conditions from 10 AM to 6 PM. Below the table, there are two verdicts: 'Outdoor verdict: GO.' and 'Beach-specific verdict: CONDITIONAL GO' followed by a detailed explanation. A list icon is visible in the bottom right corner of the app interface.

Hour	Classification
10 AM	PARTLY CLOUDY
11 AM	PARTLY CLOUDY
12 PM	PARTLY CLOUDY
1 PM	PARTLY CLOUDY
2 PM	PARTLY CLOUDY
3 PM	MIXED / INTERMITTENT
4 PM	CLEAR / MOSTLY SUNNY
5 PM	CLEAR / MOSTLY SUNNY
6 PM	CLEAR / MOSTLY SUNNY

Outdoor verdict: GO.
Beach-specific verdict: CONDITIONAL GO
because of possible fog before the target period, high humidity, and an increasingly noticeable afternoon onshore breeze. No weather-based no-go period is presently identified.

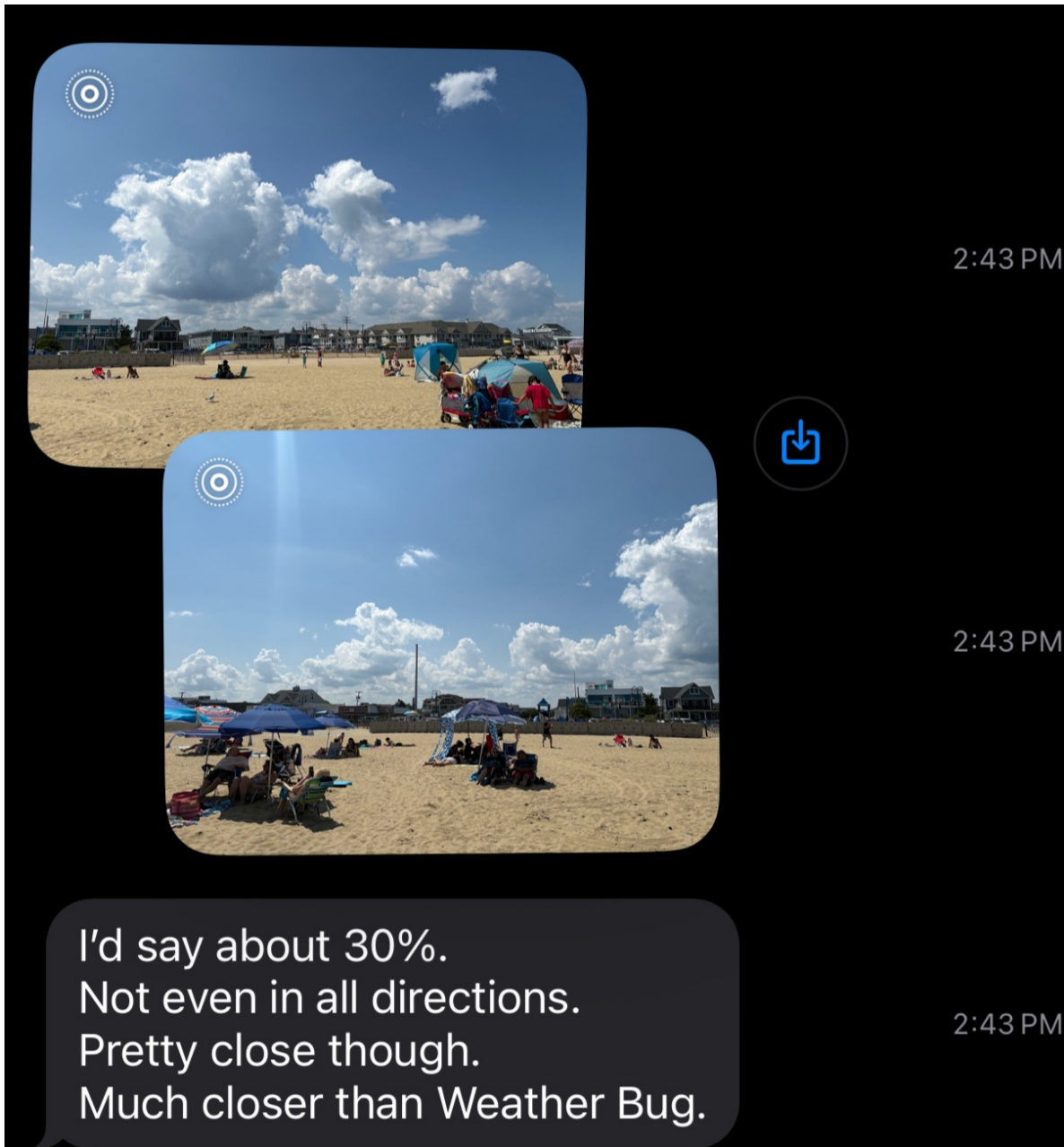
EVIDENCE HANDLING

Forecast evidence. Visible interface timestamps remain intact.

EVIDENCE PLATE 04

Plate 4 — Beach Photographs at 2:43 PM

Two contemporaneous photographs showing extensive blue sky and uneven cumulus over limited beach-facing directions.



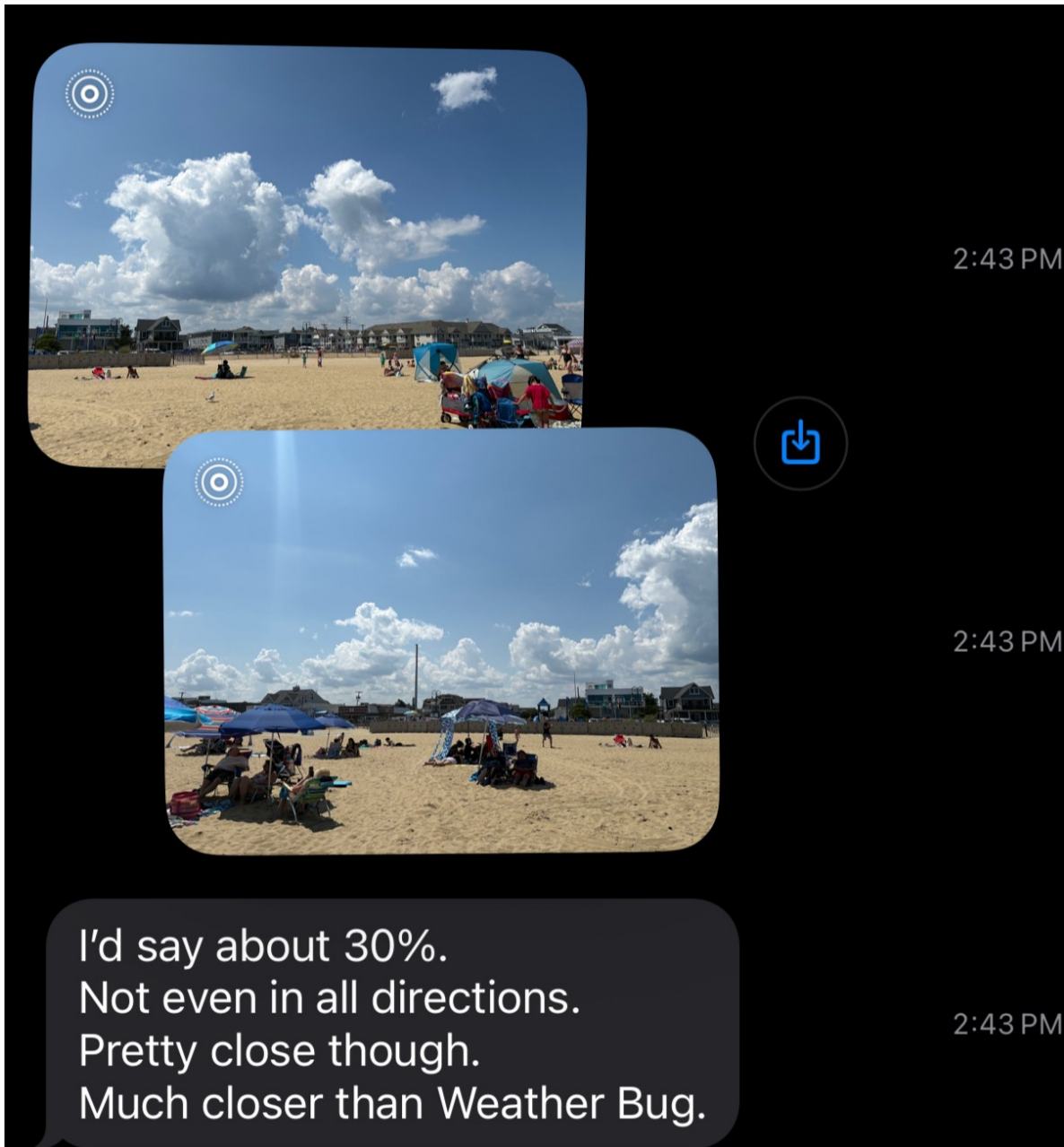
EVIDENCE HANDLING

Direct visual evidence. The image also preserves the contemporaneous observer text stating approximately 30% and noting uneven directional coverage.

EVIDENCE PLATE 05

Plate 5 — Observer Estimate at 2:43 PM

Text message stating approximately 30% cloud cover and noting that clouds were not present equally in all directions.



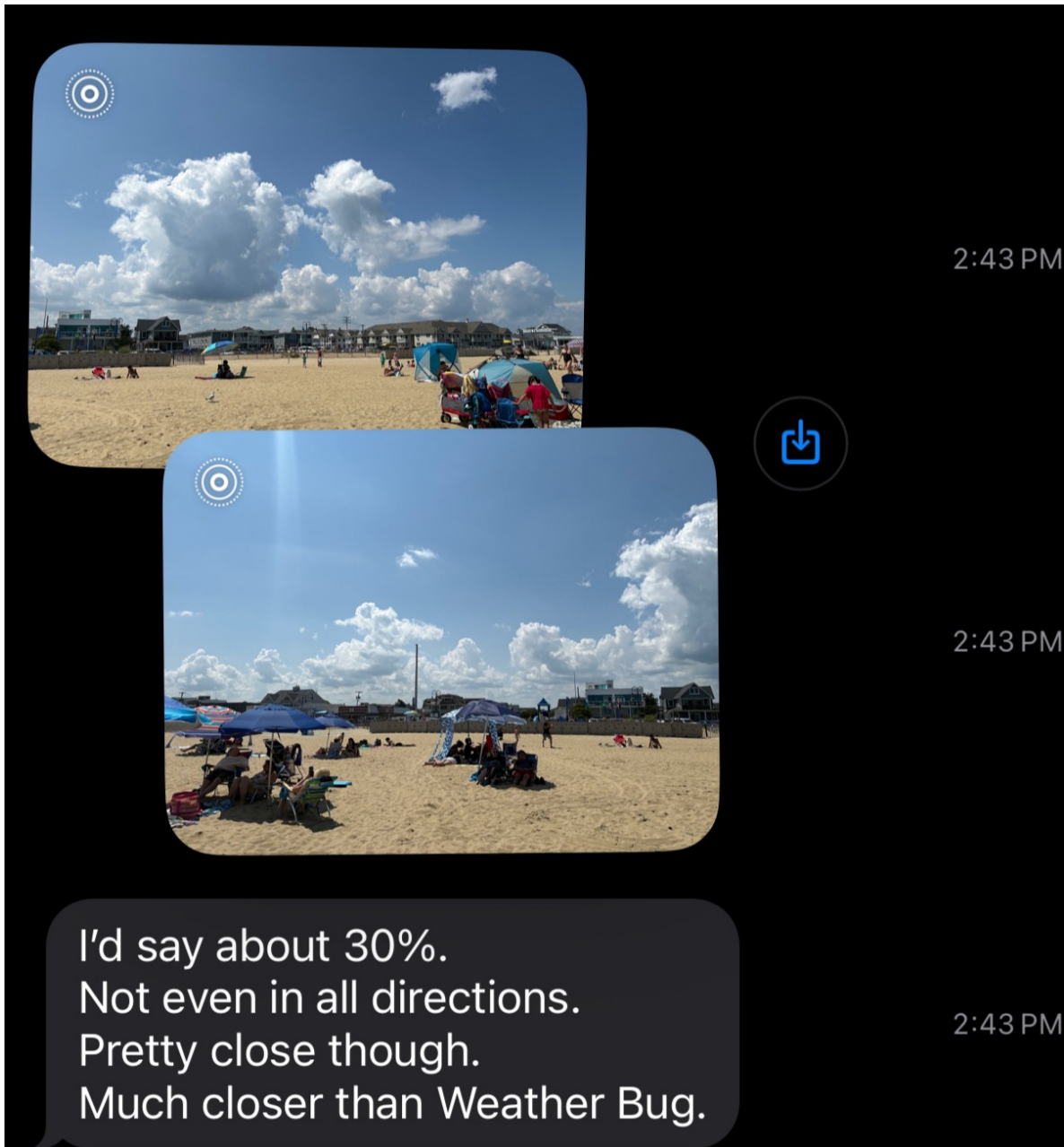
EVIDENCE HANDLING

Human estimate, not instrument measurement. Reused source image intentionally because the supplied composite contains the observer record.

EVIDENCE PLATE 06

Plate 6 — Observer Follow-Up at 2:50 PM

Text message stating approximately 40% on arrival and reporting that conditions had cleared during the preceding hour.



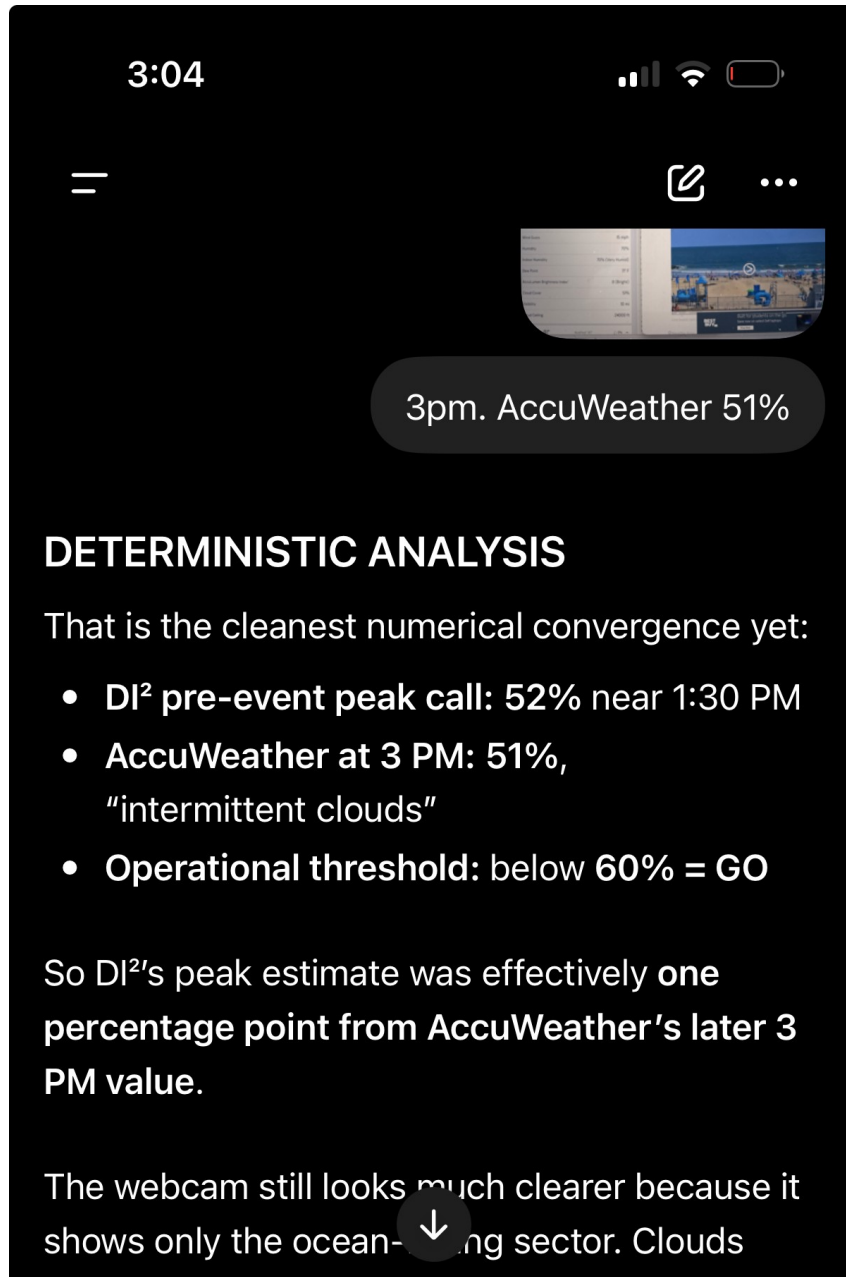
EVIDENCE HANDLING

Approximate recollection. The supplied composite preserves the contemporaneous 2:43 PM exchange; the 2:50 PM follow-up remains source text in the report.

EVIDENCE PLATE 07

Plate 7 — AccuWeather Comparison at Approximately 3 PM

Display showing 51% cloud cover and intermittent clouds.

**EVIDENCE HANDLING**

Modeled/grid product; later than the forecast peak. The supplied comparison screenshot preserves the "3pm. AccuWeather 51%" record.

EVIDENCE PLATE 08

Plate 8 — Weather Channel Display at 3:01 PM

Display showing 60% cloud cover “now,” 66% at 4 PM, 59% at 5 PM, and 27% at 6 PM.



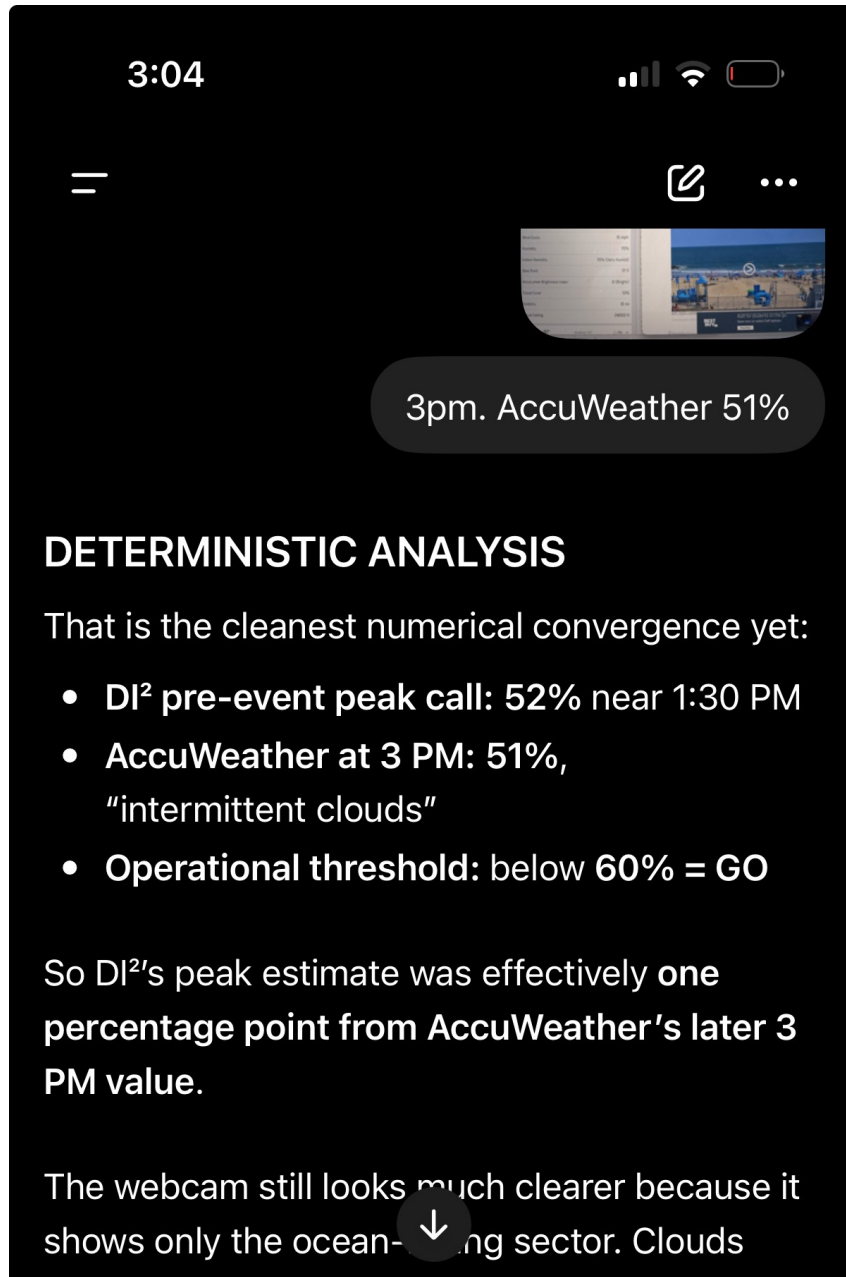
EVIDENCE HANDLING

Modeled product, not direct beach measurement. The supplied commercial display is retained without recalculation.

EVIDENCE PLATE 09

Plate 9 — Simultaneous Belmar Beach Webcam

Ocean-facing view appearing predominantly blue and sunny. Used only to characterize the visible beachfront sector.



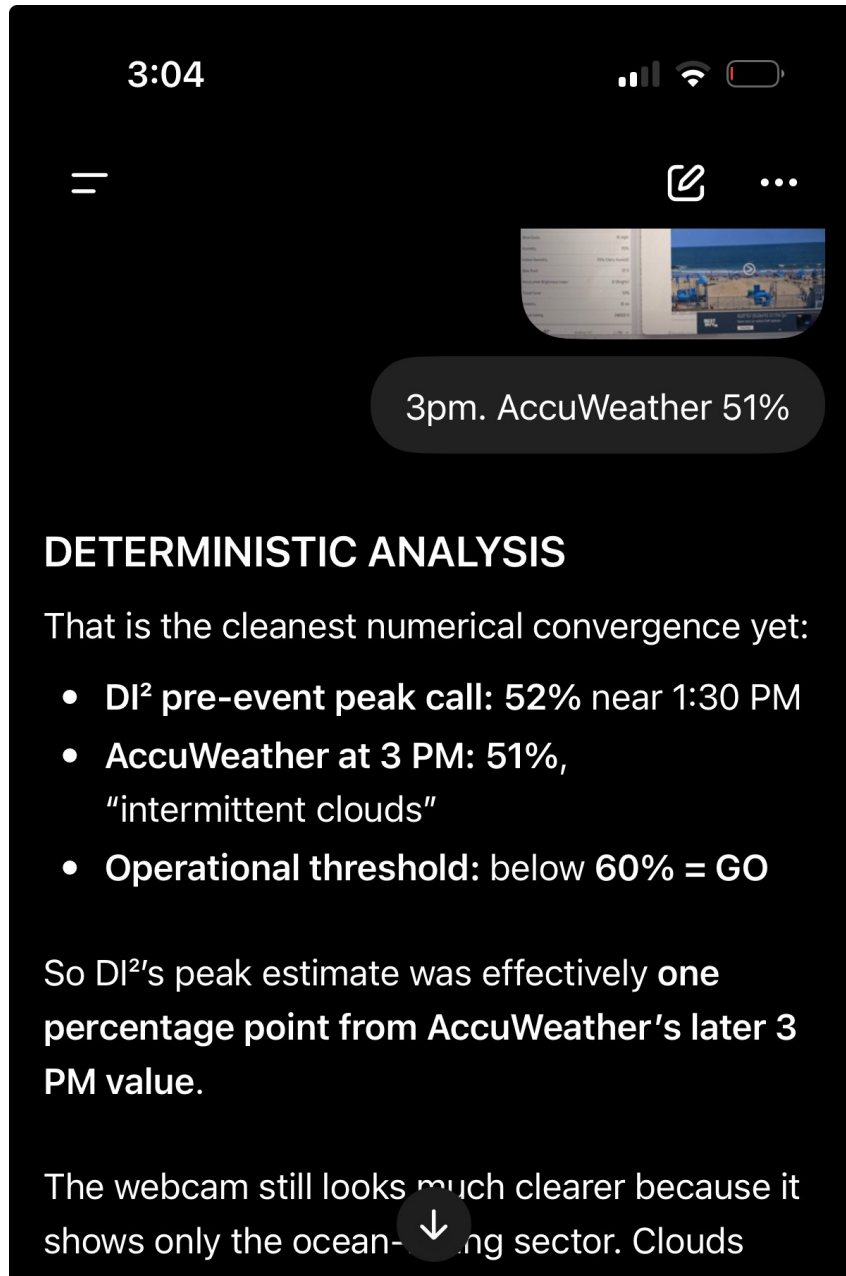
EVIDENCE HANDLING

Restricted field of view. The webcam appears as the supplied comparison screenshot's ocean-facing thumbnail.

EVIDENCE PLATE 10

Plate 10 — Derivative DI² Audit Screenshot at 3:04 PM

Later screenshot documenting the comparison between the archived 52% forecast and AccuWeather's 51% value. Not treated as independent meteorological evidence.

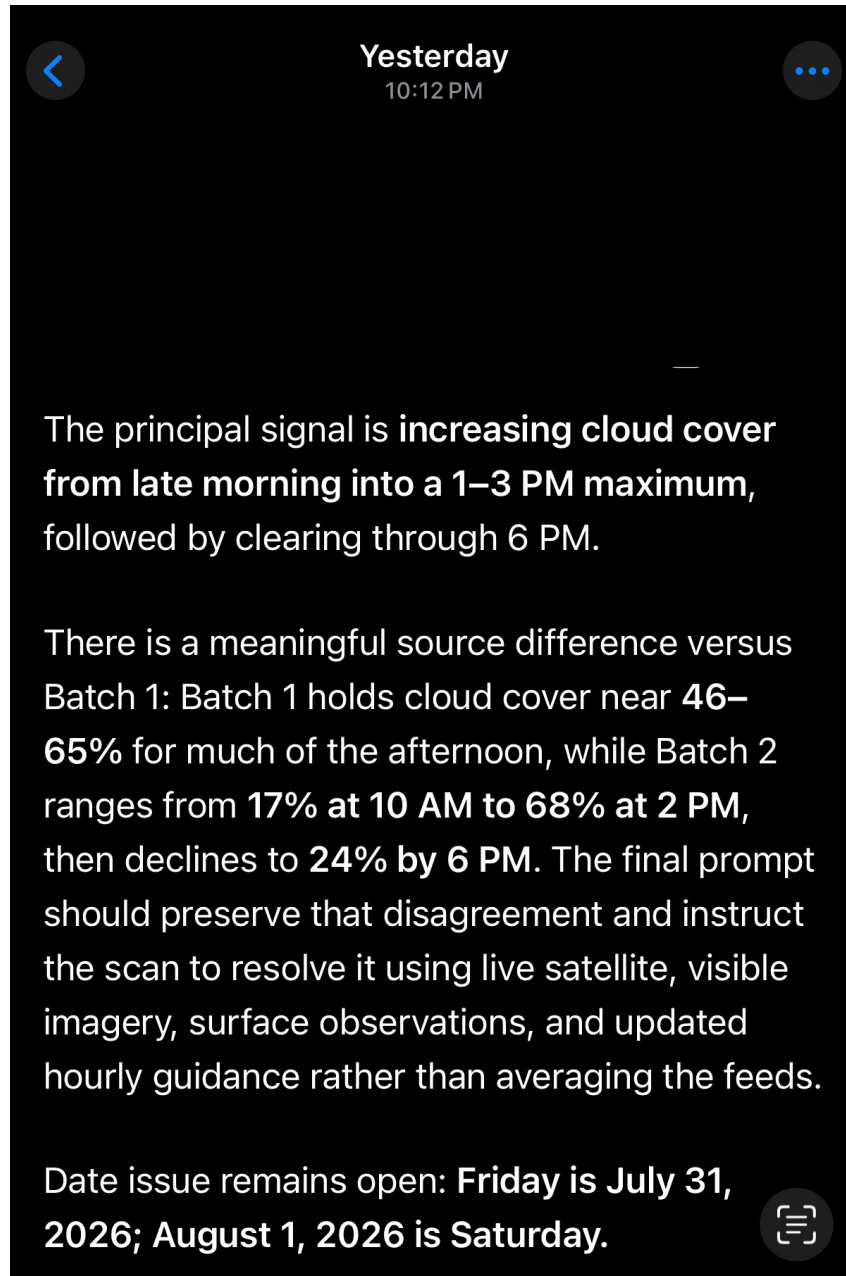
**EVIDENCE HANDLING**

Derivative analysis; not independent evidence. Reused source image intentionally because it contains the comparison record.

EVIDENCE PLATE 11

Plate 11 — Archived DI² Source-Divergence Note

Archived analysis screenshot stating that the principal signal was increasing cloud cover into a 1–3 PM maximum followed by clearing through 6 PM, and preserving the Batch 1 / Batch 2 source difference.



EVIDENCE HANDLING

Derivative analytical material. The date note is preserved as displayed and is not independently reconciled here.

EVIDENCE PLATE 12

Plate 12 — DI² Clearing and Date-UI Note

Analysis screenshot preserving the statement that the principal signal was increasing cloud cover from late morning into a 1–3 PM maximum, followed by clearing through 6 PM, together with the displayed date note.

The principal signal is **increasing cloud cover from late morning into a 1–3 PM maximum**, followed by clearing through 6 PM.

There is a meaningful source difference versus Batch 1: Batch 1 holds cloud cover near **46–65%** for much of the afternoon, while Batch 2 ranges from **17% at 10 AM to 68% at 2 PM**, then declines to **24% by 6 PM**. The final prompt should preserve that disagreement and instruct the scan to resolve it using live satellite, visible imagery, surface observations, and updated hourly guidance rather than averaging the feeds.

Date issue remains open: **Friday is July 31, 2026; August 1, 2026 is Saturday.**

EVIDENCE HANDLING

Derivative analytical material. The visible timestamp and date text remain intact.

14. Verification Block

Field	Value
Artifact-ID	SWDI2-BELMAR-CLOUD-AUDIT-20260731-001
Input-Hash	NOT COMPUTED
Output-Hash	NOT COMPUTED
Rule-Layer	Tier-18 weather-audit posture
Entropy	$\Delta H \leq 0.0041$ declared; not independently measured
Replay status	Source-bound canonical report
Verification status	Evidence chronology reviewed; exact numerical peak unresolved
Hash scope	Not established

15. Deterministic Evidence Receipt

Receipt field	Value
Report-generation timestamp	July 31, 2026, 3:07 PM EDT
Input classification	Post-event weather decision audit
Evidence classes used	A, B, C, D, E, F, G

Constraint stack

Pre-event archive preservation → timestamp ordering → evidence-class separation → observation/forecast distinction → threshold analysis → exact-value restraint → limitations disclosure

Unresolved facts

- exact whole-sky cloud fraction at 1:30 PM;
- exact maximum cloud fraction during the event;
- whether the 1:56 PM station category corresponded to a value below or above 60%;
- full-sky conditions outside the photographed and webcam sectors.

Replay eligibility: Eligible when the same preserved images, timestamps, observer statements, and classification rules are used.

FINAL AUDIT STATUS

OPERATIONAL DECISION VERIFIED — PEAK TIMING SUPPORTED — EXACT PEAK PARTIALLY VERIFIED